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# Digital Assignment - 1 :

## (1) Implementation of Cluster computing in VIT Chennai :

VIT University can implement cluster computing to improve efficiency, reliability and scalability in academic, research and administrative operations. Cluster computing refers to a collection of interconnected computer (nodes) that work together as a single system to handle computational tasks.

### Types of cluster systems :

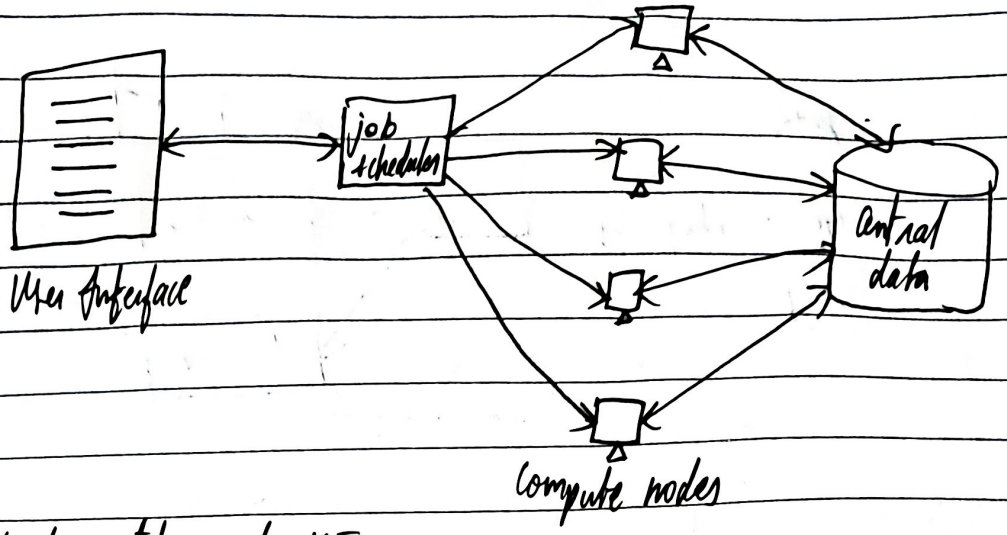
i) High-performance Computing (HPC) clusters :

- Purpose : HPC clusters enable VIT to support computationally intensive tasks such as scientific simulations, deep learning and big data processing

- How it works : Multiple nodes work in parallel, connected via high speed interconnects like InfiniBand or Gigbit Ethernet. Tasks are divided into smaller sub tasks and executed across multiple computing nodes.

- Message passing interface (MPI) and OpenMP are frameworks commonly used for communication between nodes

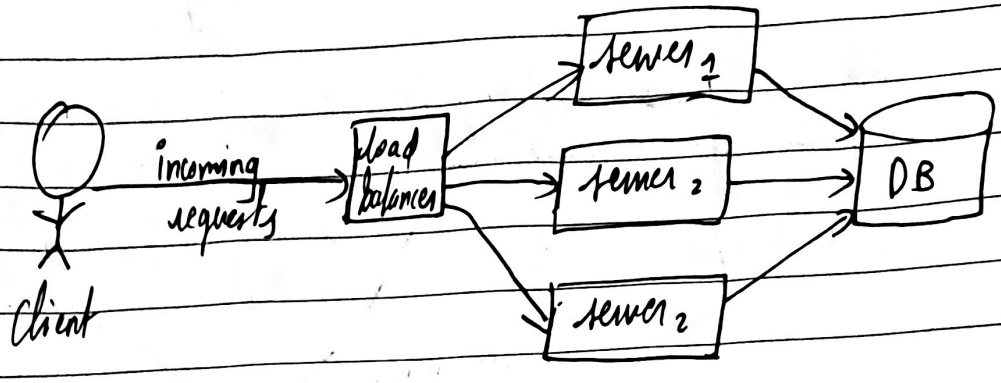
- Use cases in VIT :
  - Bioinformatics : Genome sequence & molecular simulations
  - Physics & engineering simulations : computational fluid dynamics, finite element analysis.
  - AI & ML : training largescale model on GPU.




- Implementation at VIT:
  - Hardware: high performance GPUs, CPU clusters, infiniband
  - Software: Slurm (job scheduling), OpenMPI, Tensorflow for AI workloads.

ii) Load-balancing clusters:

- Purpose: Ensures reliability, scalability for VIT's online services, including student portals, e-learning systems and admission portals.
- How it works: Requests from users are distributed across multiple servers.
- Load balancers prevent any single server from being overwhelmed, ensuring continous availability.
- Use cases at VIT:
  - FFCS: Slot booking servers' load is balanced during peak
  - Online admissions: prevents downtime during peak application
- Implementation:
  - hardware: Multiple web servers, high-availability configurations
  - software: Nginx, HAProxy, MySQL Cluster for database replication.



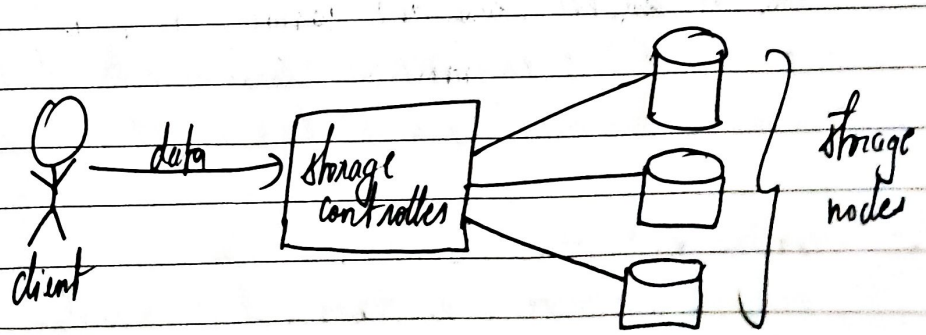

iii) Storage clusters:

- Purpose: Manages large-scale student and research data storage securely & efficiently.

- How it works: Data is distributed across multiple storage nodes for high availability and fault tolerance.

- Use cases of VIT:
  - Research data storage: cloud-based repositories for academic papers.
  - Student record mgt: secure archival of admission & result data.
  - Backup system: disaster recovery planning

- Implementation:
  - hardware: SSD, RAID for redundancy
  - software: Ceph, HDFS for file storage



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② Leveraging cloud computing at VIT University :

Cloud computing can help VIT University optimize its academic, research and administrative operations by offering scalable, flexible & cost effective infrastructure. The university can use different cloud models to efficiently handle e-learning platforms, research collaboration, secure data management & overall digital transformation.

Types of CC models & their implementation at VIT :

i) Public cloud :

- Public cloud services are provided by 3rd party providers like AWS, Azure, Google Cloud offering on-demand computing resources over the internet.

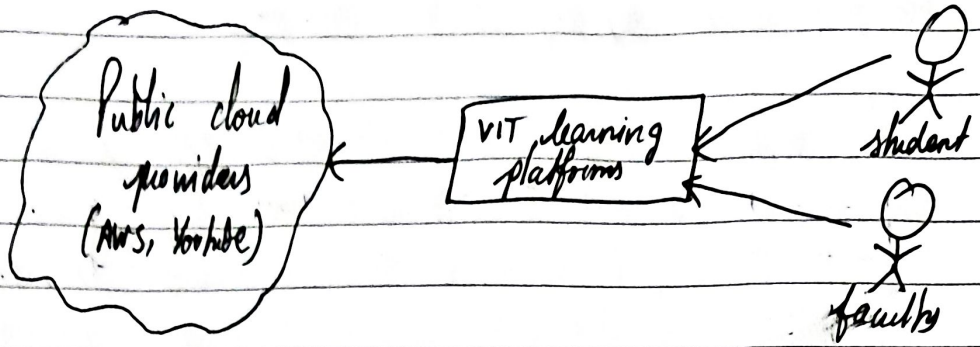
- Benifits :
  - Scalability : handles thousands of students accessing LMS simultaneously
  - cost-effectiveness : No need to maintain expensive on-premise hardware.
  - Accessibility : Students & faculty can access resources anytime, anywhere

- Use cases in VIT :
  - LMS hosting : platforms like moodle, coursera or custom e-learning portals can run on AWS.
  - Video lectures & virtual labs : hosted on youtube, AWS S3 or google drive for seamless access
  - Online exams : proctored exams can use google cloud vision AI for identity verification

ii) Private cloud :

- A private cloud is an infrastructure exclusively used by VIT university for high security & confidential data processing

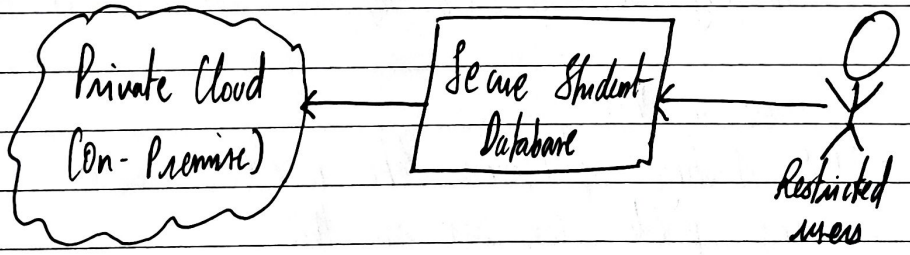
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public cloud

- Benifits:
  - enhanced security: ensures sensitive data remains private
  - data compliance: helps meets legal regulations
  - custom control: VIT can configure own cloud

- Use cases:
  - Student information system: admission details, grades, attendance.
  - Research data: securely stores proprietary research work.
  - faculty & payroll systems: keeps employee records & financial transaction secure.




iii) Hybrid cloud:

- A hybrid cloud combines both public & private cloud models, balancing cost efficiency & security.

- Benifits:
  - best of both worlds: sensitive data remains in private whereas non sensitive data transferred to public cloud.
  - cost optimization: reduces infrastructure cost.

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- seemless integration: ensures interoperability between private & public

• Use cases:
- online exam management: platform in public, results in private
- Research data storage: open access projects in public, classified work in private.

iv) Community cloud:
• A community cloudd is shared among multiple universities or research institutions providing resource sharing

• benifits:
- Encourages collaboration: Allows VIT to share computing resources with IITs, NITs & global universities
- Reduces infrastructure cost: instead of building separate research clouds, universities can pool resources.
- facilitates multi university research: scientists & students can access shared DBs, AI models etc

• Use cases:
- shared research supercomputer
- cross university data sharing

